

A Bipartite Nine-Cell Framework for Crowd Safety Analysis: Venue P-Report and Event DIM ICE as Complementary Instruments

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Abstract. This paper proposes a bipartite nine-cell framework for crowd safety analysis, consisting of two structurally analogous 3×3 matrices: a venue P-report and an event DIM ICE analysis. The two matrices share the horizontal axis (Ingress, Circulation, Egress) but differ on their vertical axis, scoring logic, and object of analysis. The venue P-report addresses static physical baseline certification through a Hardware/Software/Management vertical axis ordered by cost of change; the event DIM ICE, preserved in Still's original form, continues to address event-specific dynamic planning through the Design/Information/Management axis. The paper introduces the Non-Contradiction Principle as the central analytic tool of the venue P-report: within each column, the three cells must be mutually consistent, with three canonical failure modes (Software–Hardware, Management–Software, Management–Hardware contradictions) corresponding to identifiable classes of historical incident. A three-zone data expression (Green/Amber/Red) represents each cell's physical safety envelope, with zone boundaries anchored to phase transitions in the underlying P-index framework. Safety reports produced under the framework are reframed as productive commercial assets rather than compliance overhead, generating value through premium reduction, rent uplift, event-mix expansion, and litigation defence. Application to four historical incidents (Hillsborough 1989, Love Parade 2010, Astroworld 2021, Itaewon 2022) illustrates how each canonical failure mode maps to a pre-existing documentary contradiction that the framework's pairwise consistency check would have surfaced. The framework is positioned to complement rather than replace existing tools: DIM ICE remains the appropriate framework at the event layer, and the venue P-report addresses the gap in venue-layer certification with a compressive-asphyxia focus.

Keywords: crowd safety, DIM ICE, nine-cell framework, liability apportionment, venue certification, omission-based liability, Non-Contradiction Principle, Hardware/Software/Management, cost-of-change ordering

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1. Introduction

1.1 The Fragmentation Problem in Current Crowd Safety Practice

Contemporary crowd safety practice operates through a set of frameworks, tools, and documentation traditions that evolved to answer different questions in different eras. Fire codes such as NFPA 101 and the German MVStättVO answer the question of what a venue’s maximum static capacity should be, using egress-time calculations derived from mid-twentieth-century fire-evacuation research. Event-planning frameworks, most prominently Still’s DIM ICE, answer the question of how a specific event should be organised on a given day. Post-incident forensic reconstruction, when it occurs, is typically bespoke and case-specific.

These frameworks are individually mature within their intended scopes. They are, however, fragmented across the lifecycle of a crowd-dense venue. A venue operator holds one set of documents for licensing; an event organiser produces a different set for each event; an insurer receives a third; a court, after an incident, constructs a fourth. The documents refer to the same physical venue and the same physical people, but they use incompatible vocabularies, different analytic axes, and non-comparable output formats.

The consequence is that crowd safety, as a subject of professional analysis, lacks a stable substrate. Findings from one document cannot be transferred to another without extensive reformulation. Improvements in one layer (e.g. more rigorous event-day analysis) do not accumulate at other layers (e.g. venue certification). Historical incident data does not straightforwardly feed forward into future risk assessments because the analytic framing differs at each step.

1.2 The Dominant Framework at the Event Layer

Within the event-planning layer, the framework with the broadest global adoption is DIM ICE, developed by G. Keith Still and disseminated through the Crowd Safety and Risk Analysis training programme and its associated publications. DIM ICE analyses events across a three-by-three matrix: Design, Information, and Management crossed with Ingress, Circulation, and Egress. It is used in the United Kingdom, across the Commonwealth, in much of the Middle East, and by international event consultancies whose portfolios span multiple jurisdictions.

DIM ICE is a mature framework for its intended purpose. It provides a structured vocabulary for pre-event risk dialogue among organisers, venues, and authorities; it identifies the principal domains in which planning attention must be applied; and it has accumulated three decades of practitioner refinement. This paper does not propose an alternative to DIM ICE at the event layer. It proposes a complement at a different layer.

1.3 The Gap at the Venue Layer

The layer that DIM ICE does not address, and was not designed to address, is the venue’s static safety envelope. When an organiser applies DIM ICE to a specific event, they take the venue as given: its corridors, exits, sightlines, and capacity are inputs, not variables. But the venue itself, as a physical asset that hosts many events over its lifetime, requires a form of analysis

that DIM ICE does not provide. What is this venue’s baseline safety envelope, independent of any particular event? What are the physical limits beyond which no event can safely be run here? How does this venue’s baseline compare to others? These are venue-layer questions, and they are the questions insurers, regulators, and post-incident forensic analysts must be able to answer.

Current practice fills this gap with fire codes and per-person area rules. These are adequate for fire-evacuation planning, which is what they were designed for, but they carry no direct physical connection to compressive-asphyxia risk—the dominant mechanism in the last several decades of catastrophic crowd incidents. A venue that passes fire-code capacity requirements may nonetheless have physical characteristics that make it unsafe under certain crowd configurations, and existing frameworks provide no structured means to identify these characteristics in advance.

1.4 The Proposal of This Paper

This paper proposes a venue-domain analogue of DIM ICE. The proposed framework preserves the horizontal axis of DIM ICE (Ingress, Circulation, Egress), because this axis reflects a temporal reality that is identical for venues and events: crowds enter, move through, and exit. It replaces the vertical axis with a different set of three domains (Hardware, Software, Management) reflecting the venue’s static physical, semi-permanent informational, and dynamic operational layers. It replaces DIM ICE’s additive scoring logic (what has been done to make this event safer) with a deductive logic (what would have to be present for this venue to be unsafe, and what physical margins protect against it).

The two frameworks are structurally analogous: both are 3×3 matrices with the same horizontal axis. They are functionally complementary: the venue P-report specifies the static envelope; the event DIM ICE analyses the specific event within that envelope. They are independently usable: a venue may hold a current P-report even when no specific event is scheduled; an organiser at a pre-certified venue may prepare a DIM ICE that inherits the venue’s envelope without re-analysing it. They are jointly composable for high-attendance or high-risk events, where the two documents together constitute the full safety analysis.

The physical basis for the venue P-report’s quantitative content is the P-index framework developed in a companion paper (Tao, 2026a). The three-zone data expression used in each cell of the P-report (Green, Amber, Red) corresponds to well-defined phase transitions in the P-index framework: the Green zone corresponds to $P_{\text{baseline}} = 0$, the Amber zone to intermediate values requiring event-specific analysis, and the Red zone to (F, τ) trajectories crossing the long-duration lethal anchor of the P-index framework. This connection anchors the qualitative zone assignment to a quantitative physical basis, which is what distinguishes the present framework from a checklist and what enables its use in insurance actuarial workflows and in civil litigation.

1.5 Structure of the Paper

Section 2 introduces DIM ICE as a mature framework and identifies the scope for which it was designed and the scope beyond which it was not intended to apply. Section 3 argues that

the vertical axis of conventional nine-cell reports lacks explicit dependency structure, and that this omission is the root cause of the report-as-filling-exercise problem. Section 4 introduces the Hardware/Software/Management vertical axis and its cost-of-change ordering. Section 5 formalises the Non-Contradiction Principle as the central analytic tool of the framework. Section 6 defines the venue nine-cell matrix cell by cell. Section 7 specifies the three-zone data expression and its physical basis. Section 8 describes how the venue P-report and event DIM ICE interface. Section 9 develops the productive-capital framing that motivates commercial adoption. Section 10 applies the framework to documented historical incidents. Section 11 discusses scope limitations. Section 12 concludes.

2. DIM ICE as a Mature Framework

2.1 Origin and Development

The DIM ICE framework was developed by G. Keith Still through the 1990s and 2000s as a teaching and analysis tool within what became the Crowd Safety and Risk Analysis training tradition. Its structure—Design, Information, and Management crossed with Ingress, Circulation, and Egress—provides a nine-cell matrix within which event organisers, venue operators, and public authorities can jointly review the risk profile of a specific event.

DIM ICE emerged from practitioner engagement with the sequence of major crowd incidents that punctuated the late twentieth century, including the Hillsborough disaster of 1989 and the subsequent Taylor Report. Its analytic form reflects the lessons that professional crowd safety practice extracted from that period: that safety failures rarely reduce to a single cause, that meaningful pre-event review requires a structured vocabulary shared across stakeholders, and that the categories of analysis must span both physical planning and real-time management.

2.2 Structure and Analytic Strengths

The horizontal axis of DIM ICE—Ingress, Circulation, Egress—partitions the temporal phases of a crowd’s engagement with a venue. This partition is intuitive to practitioners and corresponds to distinct operational challenges: pre-event arrival management, in-event movement and dwell, and post-event dispersal each present different risk profiles requiring different planning attention.

The vertical axis—Design, Information, Management—partitions the modalities through which event planners exert control. Design captures the physical and procedural decisions made before the event: entry route configuration, wayfinding layout, exit assignments, dispersal routes. Information captures the communication systems that guide crowd behaviour: signage, public address, mobile notifications, staff radio protocols. Management captures the live operational decisions made during the event: staffing deployment, incident response, adjustments to conditions as they emerge.

The nine resulting cells provide a structured checklist across which pre-event review proceeds. For each cell, planners consider what has been arranged, what could go wrong, and what contingency exists. The framework’s strength lies in its comprehensiveness: an event that has been examined systematically across all nine cells is unlikely to have entire domains overlooked.

2.3 Global Adoption

DIM ICE has achieved broad international adoption, particularly in jurisdictions with significant public-event industries and established professional crowd safety practice. It is the reference framework in United Kingdom event licensing practice; it is used across Commonwealth countries with related regulatory traditions; it is applied in the Middle East in the context of large-scale religious gatherings and mega-events; and it is used by international consultancies whose portfolios span multiple jurisdictions.

Beyond direct application, DIM ICE has shaped the vocabulary of crowd safety practice. Even in venues and events where the specific nine-cell format is not used, the practitioner distinction between design, information, and management, and between ingress, circulation, and egress, has become part of the standard analytic language of the field. In this sense DIM ICE functions as more than a specific tool: it is a shared intellectual substrate for professional crowd safety analysis.

2.4 The Scope DIM ICE Was Designed For

DIM ICE was designed for event-specific pre-event planning. Its analytic form assumes an identifiable event with defined parameters: an expected attendance, a scheduled duration, a known audience profile, a specific venue. It supports the review of these parameters in combination and helps identify aspects of the event's design that require refinement before approval.

Within this scope, DIM ICE is well suited. It is qualitative, which matches the qualitative nature of most event planning decisions. It is additive, which matches the planning practitioner's task of assembling adequate provisions across many domains. It is non-comparative, which matches the reality that each event is different and that comparability across events is not the primary requirement.

2.5 The Scope DIM ICE Was Not Designed For

The scope for which DIM ICE was not designed, and does not attempt to serve, includes several tasks that have become increasingly important as the insurance, regulatory, and legal environments around crowd safety have developed. These include:

Venue-level static baseline certification. A venue exists independently of any specific event. Its physical characteristics—the width of its corridors, the placement of its exits, the topology of its transitions—form a static envelope within which many different events will be run over the venue's operational life. Analysing this static envelope requires a framework that takes the venue rather than the event as its object of analysis. DIM ICE, being event-scoped, does not address this task.

Cross-venue comparability. An insurer holding a portfolio of policies across multiple venues, or a regulator setting standards that apply across a jurisdiction, requires an analytic form whose outputs are comparable across venues. Qualitative event-planning documents do not produce comparable outputs, because each is written to the specific event and venue at hand. DIM ICE reports are not intended to be compared across venues.

Quantitative certification scoring. Certification frameworks in mature engineering sectors—structural engineering, fire safety, catastrophe insurance—rely on quantitative outputs that map to defined risk categories. DIM ICE is qualitative and produces narrative documentation rather than quantitative scores. This is appropriate for its intended use but does not meet the requirement of certification frameworks that need to distinguish among levels of achieved safety.

Post-incident liability apportionment. After an incident, civil courts and insurance claim processes need to apportion responsibility among the parties whose actions or omissions contributed to the outcome. Doing so requires an analytic framework that separates the domains under each party’s control and that quantifies the residual risk each party accepted. DIM ICE, designed for pre-event planning across a unified organiser-venue-authority partnership, does not partition responsibility in the way that post-incident analysis requires.

These scope limitations are not deficiencies of DIM ICE. They are consequences of the specific purpose for which DIM ICE was designed. The framework proposed in this paper is intended to address these scope areas without disturbing the domain in which DIM ICE remains the appropriate tool.

2.6 The Complementary Role Identified in This Paper

The framework proposed in this paper is positioned as a complement to DIM ICE, not as a replacement. DIM ICE continues to be the appropriate tool for event-layer analysis. The venue P-report proposed here is the appropriate tool for venue-layer analysis. The two frameworks share the horizontal axis (Ingress, Circulation, Egress), which allows outputs from one to interface with inputs to the other. They differ on the vertical axis and on the scoring logic, which reflects their different objects of analysis.

Practitioners familiar with DIM ICE will find the venue P-report structurally recognisable: it is another 3×3 matrix with the same horizontal axis. The vertical axis (Hardware, Software, Management) is chosen to reflect the venue’s static physical, semi-permanent informational, and dynamic operational layers, which correspond by rough analogy to Design, Information, and Management in the event context. The productive-capital framing developed in Section 9 applies to both frameworks: high-quality venue P-reports and high-quality event DIM ICE analyses each generate downstream commercial value that supports their continued rigorous production.

3. The Missing Structural Logic in Current Nine-Cell Reports

3.1 The Filling-Exercise Problem

A nine-cell report, as conventionally produced, assigns content to each of nine cells. An analyst working through the matrix considers each cell in turn: what has been designed for ingress, what information supports ingress, what management provisions exist for ingress; what has been designed for circulation, and so on through all nine cells. Once every cell contains substantive content, the report is judged complete.

This mode of production, which we term the filling exercise, has a specific structural weakness: it can produce reports that are locally complete but globally inconsistent. Each cell, examined in isolation, may describe appropriate provisions. The cells taken together may nonetheless describe strategies that mutually defeat each other.

Consider a hypothetical example. A venue’s Design cell for egress specifies three exit routes: primary through the main entrance, secondary through the east loading bay, tertiary through the north emergency stair. Its Information cell specifies fixed signage guiding crowds to the primary and secondary routes. Its Management cell specifies that on high-attendance nights, staff position portable barriers to close the loading bay to non-emergency use and route all traffic through the primary entrance.

Each cell, examined alone, is defensible. Design provides multiple routes; Information communicates the standard routes; Management responds to operational realities. But the three together describe a system in which, on the nights of highest attendance, the actual egress capacity has been reduced to one route by a management decision, while fixed signage still communicates two routes, while emergency egress capacity assumed by the venue’s licensing depended on three. A conventional nine-cell report of this venue would show three substantively populated cells in the egress column. The internal contradiction would not appear anywhere in the report.

The example is stylised, but the pattern is not. Every major post-incident investigation of the past three decades has identified inconsistencies of this general form. In each case, some element of pre-incident planning existed for each of the nine domains. In each case, the failure was in the interaction among the elements, not in the absence of any single one.

3.2 Why the Vertical Axis Is Under-Structured

The horizontal axis of a nine-cell report—Ingress, Circulation, Egress—carries a clear temporal logic. The three phases are sequential and mutually exclusive: a person is entering, or moving, or exiting; the phases correspond to distinct operational challenges; the ordering is dictated by the physical reality of the event.

The vertical axis, in both Still’s original DIM ICE and in most variants that have followed, is treated as three parallel domains without an explicit ordering. Design, Information, and Management are described as three modalities of control, each independently important. The practitioner is not directed to consider any particular relationship among them.

This treatment reflects a genuine analytic choice: the three vertical domains are not temporally sequential in the way that ingress, circulation, and egress are. Design happens before the event, but Information and Management activities interleave with each other throughout the event. There is no obvious sequential ordering to impose.

However, the absence of a temporal ordering does not mean the absence of any ordering. The three domains differ in a different way: they differ in the cost and timescale on which each can be changed. Design decisions, once committed to physical infrastructure, are the hardest to modify. Information systems, once deployed, are moderately hard to modify. Management provisions are, by definition, the layer at which real-time modification is possible. This cost-of-change ordering is present in DIM ICE implicitly, but it is not formalised, and the practitioner

reviewing a report is not directed to apply it.

The venue P-report proposed in this paper makes the cost-of-change ordering explicit, and formalises the analytic principle that follows from it. That principle—the Non-Contradiction Principle formalised in Section 5—is what converts the nine-cell matrix from a filling exercise into a conflict-detection exercise.

3.3 Preview of Consequences

Before formalising the vertical hierarchy, it is worth previewing the consequences of making it explicit. Three consequences are particularly important.

First, the report becomes auditable in a way that filling-exercise reports are not. A third-party auditor can examine each column of the matrix and verify whether the cells in that column are mutually consistent, without needing to render an independent judgment about whether any individual cell describes adequate provisions. Consistency is a structural property that can be checked by comparing documents against each other and against the physical venue; adequacy is a domain judgment that requires expertise in each specific cell.

Second, the failure modes of the venue become identifiable in a structured vocabulary. Rather than describing an incident as a general failure of planning, one can identify the specific inconsistency between specific layers: Software contradicted Hardware in this column, or Management contradicted both Hardware and Software in that column. This structured vocabulary is directly usable in insurance underwriting, regulatory review, and civil litigation.

Third, the burden of pre-event planning becomes more precise. Rather than requiring that every cell contain substantively rich content, the framework requires that each cell's content be compatible with the cells below it in the same column. This shifts effort from maximising the quantity of documentation to ensuring the coherence of the documentation that exists—which is what the historical incident record suggests is the more consequential requirement.

Section 4 formalises the vertical hierarchy. Section 5 formalises the Non-Contradiction Principle. Section 6 defines the specific content of each cell in the venue P-report.

4. The Vertical Hierarchy: Hardware / Software / Management as a Cost-of-Change Stack

4.1 The Missing Logic in Current Nine-Cell Reports

Existing crowd safety reports—including canonical DIM ICE analyses—typically populate each of the nine cells as an independent checklist. Analysts describe what has been designed, communicated, and managed for each of ingress, circulation, and egress. The horizontal axis (Ingress / Circulation / Egress) carries a clear temporal logic: crowds enter, move, and exit in sequence. The vertical axis, by contrast, is often treated as three parallel domains without an explicit dependency structure.

This treatment has a practical consequence: the report becomes a *filling exercise* rather than a *conflict-detection exercise*. Analysts confirm that each domain has been addressed, without

a structured basis for asking whether the three domains are internally consistent. Historical incidents demonstrate that the dominant failure mode is not the absence of any single domain, but the internal contradiction between them. The management response at Hillsborough was not absent; it actively opened a gate that overwhelmed the fixed physical capacity of the terrace. The signage at Love Parade was not missing; it directed converging streams into a geometrically incompatible junction. In both cases, each cell of a conventional nine-cell report could have been marked “completed” while the venue as a whole was structurally unsafe.

The vertical axis of the framework proposed in this paper is designed to make such contradictions visible. It orders the three domains not by function but by *cost of change*—that is, by the time and resource investment required to modify each layer.

4.2 The Cost-of-Change Ordering

We define the three vertical layers as follows.

Hardware refers to the physical structure of the venue: walls, doors, corridors, stairs, barriers, gate widths, floor topology. Hardware is the layer with the highest cost of change. Modifying hardware requires construction, permitting, and typically closure of the venue for extended periods. On the timescale of a single event or a single operational season, hardware is effectively fixed.

Software refers to the semi-permanent informational and instrumentation layer built onto the hardware: fixed signage, camera networks, sensor arrays, public address systems, digital wayfinding, monitoring dashboards. Software changes are cheaper than hardware changes but still require deliberate planning, procurement, and installation. Software typically updates on a timescale of weeks to months.

Management refers to the human-led operational layer: staff deployment, live decision-making, temporary signage, crowd control instructions, incident response. Management is the cheapest layer to change and operates on a timescale of seconds to hours. It is the layer at which real-time adaptation to actual crowd conditions occurs.

The cost-of-change ordering is not a claim about importance. All three layers are essential to venue safety. The ordering reflects a physical and organisational reality: when the three layers disagree, the layer that is easier to change is the one that must give way. Hardware cannot be rebuilt during an ongoing event; management can be redeployed within minutes.

4.3 Why This Ordering Matters for Safety Logic

The cost-of-change ordering has a direct implication for how safety must be maintained: **each higher layer inherits the physical constraints of the layers below it**. Software must be designed to work with the hardware it sits on; it cannot pretend the hardware is different. Management must operate within the combined constraints of hardware and software; it cannot instruct the crowd in ways that contradict either.

This inheritance flows in one direction. Hardware does not need to accommodate software; software adapts to hardware. Hardware and software do not need to accommodate management;

management adapts to both. The reverse—hardware being reconfigured to fit management preferences, or software being overridden by ad-hoc management decisions—is precisely the failure mode that produces catastrophic incidents.

This gives the vertical axis a testable structure. For each of the nine cells, one can ask: is the content of this cell consistent with the cells below it in the same column? If a Software cell describes a signage system that directs egress through a corridor that the corresponding Hardware cell identifies as under-capacity, the two are in contradiction. If a Management cell describes a crowd-flow protocol that requires temporary blocking of a route that the Hardware cell identifies as a designated emergency egress, the two are in contradiction.

These contradictions are the objects that the framework is designed to detect.

4.4 Relationship to Still's Original DIM ICE

Still's original DIM ICE framework uses Design / Information / Management as its vertical axis. Although not explicitly stated in the standard training materials, this axis carries a similar implicit ordering: Design decisions are made before the event and are difficult to reverse once venues are booked and layouts committed; Information systems are prepared in the weeks before an event; Management is executed in real time. The temporal and cost-of-change ordering is the same.

The present framework makes this ordering explicit and applies it to the venue domain, where the cost-of-change gradient is even sharper (hardware modifications may take years). We view this as a natural extension of the logic that Still's framework already contains, not a departure from it. The two frameworks share the same underlying principle: safety analysis proceeds from what cannot easily be changed to what can.

5. The Non-Contradiction Principle

5.1 Statement of the Principle

For any venue nine-cell report, and for any column (Ingress, Circulation, or Egress), the three cells in that column must be internally consistent. Specifically:

The Software cell must not describe any provision that contradicts, undermines, or is impossible to execute under the Hardware cell of the same column.

The Management cell must not describe any provision that contradicts, undermines, or is impossible to execute under either the Hardware or Software cell of the same column.

We call this the Non-Contradiction Principle. It is the central analytic tool that distinguishes the present framework from filling-based checklist approaches.

5.2 Three Canonical Failure Modes

The Non-Contradiction Principle identifies three canonical failure modes, each corresponding to a specific downward violation.

Failure Mode I: Software contradicts Hardware. A fixed signage system, camera placement, or sensor network is deployed in a way that assumes a physical layout different from what actually exists. A common form is fixed wayfinding signage that directs pedestrian flow toward a corridor with insufficient capacity, or emergency signage that points toward a route that has been permanently modified since the signage was installed. Because the software layer is longer-lived than any single event, this failure mode tends to persist across many events and accumulates risk.

Failure Mode II: Management contradicts Software. Real-time operational decisions override the fixed informational infrastructure. A common form is the deployment of temporary signs, verbal instructions, or barrier placements that contradict the fixed signage the crowd is already responding to. Crowds encountering conflicting instructions do not average them; they fragment, and fragmentation is one of the physical precursors to the deep force chains identified in the P-index framework.

Failure Mode III: Management contradicts Hardware. This is the most severe failure mode and is present in a disproportionate share of documented catastrophic incidents. It occurs when operational decisions actively defeat the physical safety margins built into the venue. The two most common sub-forms are: (a) blocking or repurposing designated emergency egress routes for crowd-management convenience, and (b) opening or redirecting flows into areas whose hardware capacity is already at or above safe limits.

The Hillsborough incident of 1989 exemplifies Failure Mode III in its acute form. The physical hardware—pens on the terrace—had a well-defined capacity. The management decision to open Gate C, taken to relieve pressure outside the ground, delivered a large additional volume of people into terrace sections that were already at or near capacity. The management layer’s response to one problem (external crowding) actively defeated the hardware layer’s designed limit (terrace capacity). Every cell of a conventional DIM ICE report on that day could have been marked “addressed”; the failure was in the interaction between them.

5.3 Detection as a Cell-Pair Analysis

The Non-Contradiction Principle transforms nine-cell analysis from a filling exercise into a pairwise consistency check. For each of the three columns, there are three cell pairs to examine:

- Software vs. Hardware
- Management vs. Software
- Management vs. Hardware

This yields nine pairwise checks across the three columns. Each check produces a binary outcome: consistent or contradictory. Contradictory pairs are the objects of remediation.

The pairwise structure has an important property: it is *auditable by a third party without requiring domain expertise in every cell*. An auditor need not evaluate whether a particular signage system is well-designed in isolation; they need only evaluate whether the signage described in the Software cell is compatible with the hardware layout described in the Hardware cell. This compatibility check can be executed by comparing documents, walking the venue, and confirming that stated Software provisions are physically realisable on the stated Hardware, and that stated Management provisions do not conflict with either.

5.4 Relationship to the Three-Zone Data Expression

The three-zone data expression (Green / Amber / Red), introduced in Section 7, applies within each cell. The Non-Contradiction Principle applies between cells. The two operate at different levels of the same analysis: the three zones tell you the physical safety envelope of each cell in isolation; the Non-Contradiction Principle tells you whether the three cells in a column combine into a coherent safety strategy.

A venue can have every cell independently in the Green zone and still fail the Non-Contradiction Principle if the three green cells describe strategies that mutually defeat each other. Conversely, a venue with cells in the Amber zone can be safely operated if the three cells combine coherently and the residual risk is documented and delegated to the event-layer DIM ICE analysis.

5.5 Legal and Insurance Implications

The Non-Contradiction Principle has direct implications for post-incident liability analysis. When an incident occurs, existing frameworks tend to identify multiple contributing causes without a structured account of which layer failed and how the layers interacted. The Non-Contradiction Principle provides such a structure. Each identified contradiction is a specific, documentable failure of downward compatibility, and each corresponds to a specific decision-making layer with an identifiable owner.

This has two consequences. First, liability apportionment becomes more precise: the party responsible for the layer at which the contradiction originated bears the corresponding share of responsibility. Management-originated contradictions are attributable to operational leadership; software-originated contradictions are attributable to venue operators; hardware-level failures are attributable to the original design or subsequent modification. Second, the standard of care becomes clearer: any competent nine-cell analysis executed in advance of the incident could have identified the contradiction. Failure to detect a pre-existing contradiction is therefore itself a documentable omission, which links directly to the omission-based liability framework developed in later sections of this paper.

5.6 Evidentiary Posture Under the Framework

The preceding subsection identified, in general terms, that the Non-Contradiction Principle changes the analytic surface available in post-incident liability review. This subsection develops the point in more operational form. Its purpose is to describe the differences in the physical

evidentiary record that arise across four configurations of P-report holding, and to identify how those differences interact with the burden-of-proof threshold defined in the physical framework (Tao, 2026a).

5.6.1 *Scope of this subsection*

This subsection does *not* predict litigation outcomes. Predicting whether a party will prevail in any specific proceeding requires integration of the physical evidentiary record with variables that lie entirely outside the scope of a physical framework: the jurisdiction’s procedural rules and standards of proof, applicable statute and case law, precedent within the relevant appellate tradition, the composition and instructions of the trier of fact, the litigation strategies actually pursued, and the credibility judgments made by decision-makers. The framework proposed in this paper is a physical instrument; it produces evidence, not verdicts.

What the framework does describe is the *evidentiary posture* of each party under each configuration: which physical evidence each party can adduce, which evidence the opposing party can invoke against them, and where the burden of positive demonstration falls under the framework’s own internal logic (Section 4.6 of the physical framework). Practitioners in specific jurisdictions can then combine this evidentiary description with the jurisdiction-specific variables above to form their own outcome estimates. This division of labour is deliberate: the framework establishes what is physically documentable, and legal professionals establish what is legally consequential.

Four configurations are distinguished, arising from two independent binary conditions: whether the venue holds a current P-report, and—if it does—whether Red-zone cells identified in the report have been remediated. The four configurations are summarised in Table 1 and developed in the following subsection.

5.6.2 *The four evidentiary configurations*

Configuration I: Neither party holds a P-report. This is the baseline configuration and describes current practice in most jurisdictions and most venue classes. Neither the plaintiff nor the defendant can adduce a physically grounded envelope of the venue’s compressive-asphyxia risk. The evidentiary record consists of witness testimony, video reconstruction, general expert opinion on crowd dynamics, and—if any exists—compliance documentation produced under fire codes or per-person-area regulations. Because these documents do not carry direct physical connection to compressive-asphyxia risk (Section 3), they are peripheral rather than central to the physical question in dispute. The evidentiary posture is symmetric: both parties argue from indirect evidence, and the outcome depends on which side’s indirect evidence the trier of fact finds more persuasive.

Configuration II: Plaintiff invokes an independently produced P-report; defendant holds none. A distinct sub-case of Configuration I arises when the plaintiff commissions a post-incident P-report reconstruction of the venue and adduces it in litigation, while the defendant venue has no equivalent document. The framework supports such retrospective reconstruc-

	Venue holds no P-report	Venue holds a current P-report
No Red-zone cells present, or all Red-zone cells remediated	Config. I: Baseline configuration. Neither party has physical evidence of the compressive-asphyxia envelope. Both parties rely on non-physical evidence (witness testimony, video, general expert opinion).	Config. III: Defendant-favourable. Defendant can adduce the P-report as documentary evidence of a physically grounded standard of care met at the time of the incident. Plaintiff must locate the alleged failure elsewhere in the causal chain.
Red-zone cells present but unremediated	(Not distinguishable from Config. I in evidentiary terms, since no Red-zone identification exists)	Config. IV: Defendant-adverse. The P-report itself documents unremediated Red-zone cells, converting the report from a defence instrument into a positive record of pre-incident awareness.

Table 1: The four evidentiary configurations that arise from the two independent conditions of P-report holding and Red-zone remediation status. Configuration II, in which the plaintiff invokes an independently produced P-report against a defendant who does not hold one, is a variant of Config. I from the defendant’s perspective and is discussed separately in the text.

tions: the P-index computation depends on physical quantities (venue geometry, occupancy, flow patterns) that can be reconstructed post-hoc from surveillance video, structural drawings, and witness accounts, as demonstrated in the historical incident analyses of Section 10 and in the anchor framework (Tao, 2026b). The evidentiary posture in this sub-case is asymmetric in a specific direction: the plaintiff can adduce a physically grounded envelope showing where the venue fell in relation to the burden-of-proof threshold at the time of the incident, while the defendant can only respond with either (i) an alternative P-report reconstruction disputing the plaintiff’s, or (ii) non-physical evidence attempting to explain why the physical framework should not apply. The framework does not assert that response (i) is impossible; competing reconstructions are legitimate and the sensitivity analyses of the underlying physical framework (Tao, 2026a) explicitly permit range estimates. But response (ii) faces the difficulty that the venue’s baseline safety envelope is precisely what compressive-asphyxia risk analysis is designed to characterise, and disputing its applicability requires an alternative physical account that the defendant, absent their own P-report, is not positioned to supply.

Configuration III: Defendant holds a current P-report with no unremediated Red-zone cells. This is the configuration the framework’s productive-capital logic (Section 9) is designed to produce. The defendant can adduce, as documentary evidence, a physically grounded certification of the venue’s static safety envelope produced before the incident by a methodology transparent to opposing experts. The report demonstrates, within the framework’s calibrated uncertainty, that at the time of the incident the venue’s baseline P-index sat below the burden-of-proof threshold across all monitored sub-regions, or that Amber-zone cells were appropriately documented and their event-specific mitigations handed off to the event-layer DIM ICE analysis (Section 8).

Under this configuration the burden of proof shifts in the manner formalised in Section 4.6 of the physical framework: the venue has met a documented physical standard of care, and the

plaintiff’s case must therefore either (i) demonstrate that the incident arose from a cause outside the P-report’s specified domain (e.g., a non-compressive mechanism, an emergent failure of the event-layer overlay, or a documented deviation from the report’s stated Management provisions), or (ii) challenge the report’s methodological validity or its accurate execution. The framework does not immunise the defendant—plaintiffs can and do prevail on grounds (i) or (ii)—but it removes the entire class of arguments that would attribute the incident to the venue’s baseline physical envelope.

Configuration IV: Defendant holds a current P-report with unremediated Red-zone cells. This is the configuration in which the P-report converts from defence instrument to adverse documentary evidence. The report itself, produced by the defendant and held on file at the time of the incident, documents that specific cells of the venue matrix sat in the Red zone—that is, that the reconstructed (F, τ) trajectories under $P_{\max}$ scenarios crossed the framework’s long-duration lethal anchor (see Section 7). The defendant is not merely alleged to have operated an unsafe configuration; the defendant is documented to have known.

The evidentiary consequence is that the plaintiff need not construct pre-incident knowledge as an inferential matter; the defendant’s own document supplies it. The defendant’s available responses reduce to: (i) demonstrating that remediation was in progress and would reasonably have been completed but for the incident’s timing; (ii) demonstrating that the specific Red-zone cell was not causally connected to the incident (the incident occurred at a Green- or Amber-zone cell instead); (iii) challenging the calibration under which the Red-zone assignment was made. Each of these responses is available in principle; none of them removes the underlying documentary posture, which is that the defendant produced a physically grounded assessment identifying a known risk and did not close the risk before the incident occurred.

5.6.3 The self-incrimination structure of Configuration IV

The asymmetry between Configurations III and IV is the framework’s most consequential result for professional practice, because it changes what it means to hold a P-report. Under conventional compliance documentation, purchasing a report is functionally equivalent to obtaining a defence instrument: the document’s existence is the evidentiary contribution, and its contents matter only insofar as they support the defence. The framework proposed here does not permit this reduction. Because the P-report’s outputs are quantitative, physically grounded, and calibrated against a transparent methodology, its contents cannot be selectively invoked: a party holding the report holds all of its findings, including findings adverse to that party’s position in subsequent litigation.

The practical implication is that adoption of the framework does not benefit venues that adopt it in name only. A venue that commissions a P-report, receives findings identifying Red-zone cells, and neither remediates the cells nor decommissions the configurations of use that engage them, has produced a documentary record of unaddressed known risk. In the language of tort practice this is the paradigmatic condition under which the standard of care escalates: the party is not asked what a reasonable operator would have known, because the party’s own document records what this operator did know.

This is why the productive-capital framing of Section 9 is not merely an economic argument but a structural one. The value channels described in that section—premium reduction, rent uplift, event-mix expansion, litigation defence—all depend on the venue closing Red-zone cells that the report identifies. A venue that captures the marketing value of holding a report while leaving Red-zone cells open captures the value only until an incident occurs, at which point the report inverts from asset to liability. The framework’s economic logic and its evidentiary logic point in the same direction: adoption without remediation is worse than non-adoption, and the two conditions of Configuration IV are not independently avoidable.

The four configurations together describe the physical evidentiary structure that the framework produces. What follows from this structure in any specific proceeding—which arguments prevail, which apportionment is reached, which precedents are set—is a matter for practitioners in the jurisdictions where such proceedings occur. The framework’s contribution ends at documenting what is physically the case; the rest is legal practice.

6. The Venue Nine-Cell Matrix

This section specifies the content of each of the nine cells in the venue P-report. Each cell is defined by four elements: (i) what the cell measures, (ii) which physical or organisational quantities are used to populate it, (iii) the boundary conditions between Green, Amber, and Red zones, and (iv) typical failure modes to which the cell is susceptible. The three-zone data expression and its physical basis are developed in detail in Section 7; this section focuses on cell-by-cell specification.

Figure 1 shows the venue P-report alongside the event DIM ICE for comparison. The two matrices share the horizontal axis (Ingress, Circulation, Egress) but differ on the vertical axis. The interface between them is developed in Section 8.

6.1 Row 1: Hardware

The Hardware row addresses the venue’s static physical infrastructure as it bears on each of the three temporal phases. Its content changes only when the venue is physically modified.

6.1.1 *Hardware × Ingress*

What the cell measures. The physical capacity of the venue’s entry infrastructure: number and width of ingress points, queue accumulation space upstream of each, screening or ticketing throughput, and the geometric compatibility of the entry configuration with expected arrival patterns.

Quantities used. Per-gate throughput (persons per minute per metre of gate width), total ingress capacity per unit time, and the ratio of queue accumulation space to peak arrival rate. Where P-index calculation is applied, the local P_{baseline} and P_{max} under standardised arrival scenarios.

Zone boundaries. Green corresponds to $P_{\text{baseline}} = 0$ under design-load arrival scenarios. Amber corresponds to $P_{\text{baseline}} = 0$ under design load but $P_{\text{max}} > 0$ under stress scenarios. Red

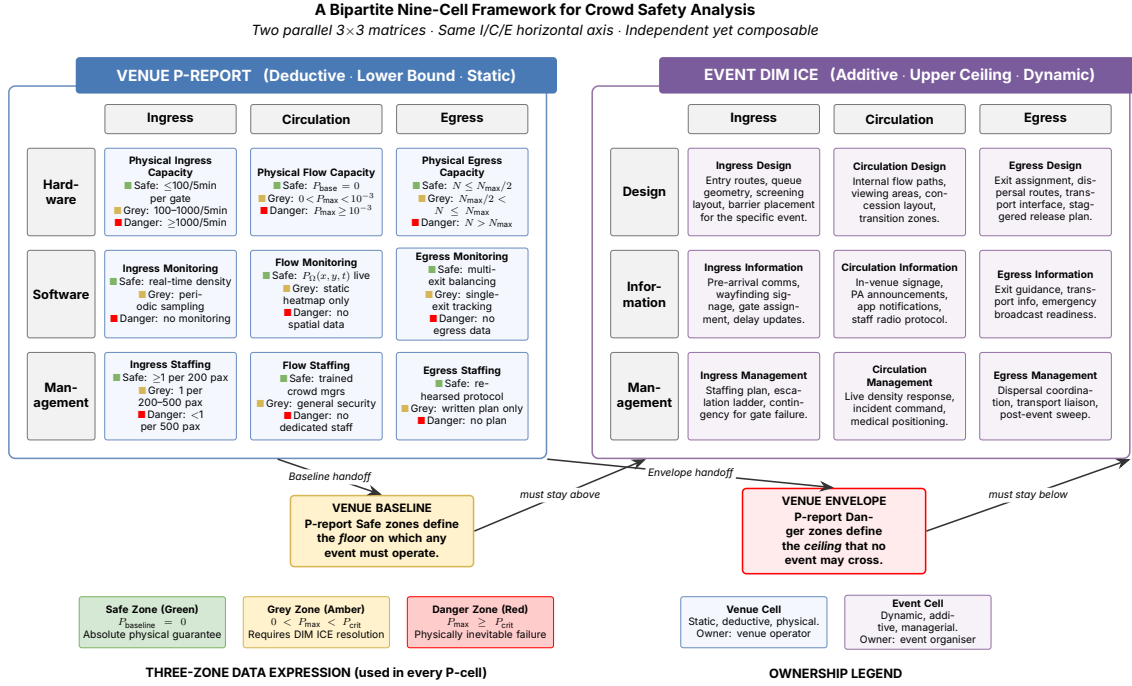


Figure 1: Venue P-Report (left) and Event DIM ICE (right) share the Ingress/Circulation/Egress axis but differ in vertical dimension, scoring logic, and ownership. The interface layer defines how the two matrices compose.

Figure 1: The bipartite nine-cell framework: venue P-report (left) and event DIM ICE (right) share the Ingress/Circulation/Egress horizontal axis but differ in vertical structure, scoring logic, and ownership. The interface layer defines how the two matrices compose. A full-resolution standalone version (22×12.375 inches) is available by compiling `figures/fig_dual_matrix.tex` independently.

corresponds to $P_{\text{baseline}} > 0$ even under normal arrival patterns.

Typical failure modes. Insufficient gate width for peak arrival rate; queue accumulation space geometry that allows crowd wrapping and cross-flow; single-point-of-failure gate configurations without physical redundancy.

6.1.2 Hardware $\times$ Circulation

What the cell measures. The physical capacity of internal circulation: corridor widths, transition geometry, sightline configuration, permanent barrier placement, and the ratio of circulation space to occupancy load.

Quantities used. Corridor throughput capacity, geometric parameters of transitions (radius, angle), permanent-obstacle inventory, and $P_{\text{baseline}}/P_{\text{max}}$ under distributed-occupancy scenarios.

Zone boundaries. Green corresponds to $P_{\text{baseline}} = 0$ across all monitored sub-regions P_{Ω} under design occupancy. Amber corresponds to Green globally but Amber-band values in identified sub-regions. Red corresponds to $P_{\text{baseline}} > 0$ in any sub-region under design occupancy.

Typical failure modes. Bottleneck geometry at internal transitions; corridors that narrow in the direction of expected flow; permanent obstacles that reduce effective width below design specification.

6.1.3 Hardware $\times$ Egress

What the cell measures. The physical capacity of the venue's exit infrastructure: number, width, and distribution of egress points; dispersal geometry downstream of each; and compatibility of the egress configuration with the venue's licensed occupancy.

Quantities used. Per-exit egress rate, total egress capacity per unit time, distribution of egress points relative to occupancy centroid, N_{max} (the venue's physical capacity limit at which P_{baseline} transitions from zero), and comparison of N_{max} to licensed capacity.

Zone boundaries. Green corresponds to N_{max} exceeding licensed capacity by a designated safety margin, with $P_{\text{baseline}} = 0$ at licensed capacity. Amber corresponds to N_{max} approaching licensed capacity. Red corresponds to N_{max} below licensed capacity.

Typical failure modes. Insufficient egress width for licensed occupancy; egress concentration on one side of the venue when occupancy is distributed; egress routes that converge into insufficient downstream capacity.

6.2 Row 2: Software

The Software row addresses the semi-permanent informational and instrumentation layer built onto the hardware. Its content changes on a timescale of weeks to months.

6.2.1 Software $\times$ Ingress

What the cell measures. The information systems that guide crowds through the ingress process: fixed signage, wayfinding, digital pre-arrival communications, and monitoring instrumentation at entry points.

Quantities used. Coverage of ingress zones by fixed signage; completeness of digital wayfinding; real-time monitoring resolution and latency at entry; and compatibility of the software layer with the ingress hardware described in the corresponding Hardware cell.

Zone boundaries. Green corresponds to full signage coverage consistent with hardware, real-time density monitoring at all gates, and integrated pre-arrival communications. Amber corresponds to complete coverage but with periodic (rather than continuous) monitoring or partial digital wayfinding. Red corresponds to signage gaps, monitoring gaps, or software-hardware contradictions.

Typical failure modes. Fixed signage that directs to routes no longer available; monitoring cameras with blind spots at critical ingress geometry; digital wayfinding inconsistent with fixed signage.

6.2.2 Software $\times$ Circulation

What the cell measures. The information systems that guide crowd behaviour during in-venue movement: internal signage, public address coverage, spatial monitoring instrumentation, and any dynamic display or notification systems.

Quantities used. Coverage of internal circulation zones by fixed signage and PA; spatial and temporal resolution of $P_{\Omega}(x, y, t)$ monitoring; latency between observation and analyst response; and compatibility with the hardware layout.

Zone boundaries. Green corresponds to live $P_{\Omega}(x, y, t)$ heatmap availability with sub-minute latency across all monitored zones. Amber corresponds to static or delayed monitoring with periodic sampling. Red corresponds to absence of spatial monitoring data.

Typical failure modes. Sightline-blocked cameras; PA dead zones in critical circulation areas; monitoring systems whose output is not integrated into any operational decision loop.

6.2.3 Software $\times$ Egress

What the cell measures. The information systems that support egress: exit signage, emergency broadcast readiness, exit-load monitoring, and transport-interface information systems.

Quantities used. Coverage of egress routes by fixed signage; readiness of emergency broadcast systems; monitoring capacity for balanced multi-exit egress; and integration with transport-interface information.

Zone boundaries. Green corresponds to complete egress signage, tested emergency broadcast readiness, and real-time multi-exit load balancing. Amber corresponds to complete signage but single-exit or delayed monitoring. Red corresponds to signage gaps, untested emergency

systems, or absence of egress monitoring.

Typical failure modes. Emergency signage that points to modified routes; PA systems untested under load; monitoring that does not distinguish among egress channels.

6.3 Row 3: Management

The Management row addresses the human-led operational layer: staffing, live decision-making, temporary interventions, and incident response. Its content changes on a timescale of seconds to hours.

6.3.1 Management $\times$ Ingress

What the cell measures. The human resources and procedures that operate the ingress process: staffing levels, training standards, escalation ladders, and contingency provisions for ingress-phase incidents.

Quantities used. Ratio of trained ingress staff to expected peak arrival rate; documented escalation protocols; and compatibility of staffing provisions with the ingress hardware and software.

Zone boundaries. Green corresponds to at least one trained ingress steward per two hundred expected arrivals per hour, with documented escalation. Amber corresponds to staff ratios between one per two hundred and one per five hundred. Red corresponds to staffing ratios below one per five hundred or absence of documented escalation.

Typical failure modes. Understaffing during peak periods; general security personnel without ingress-specific training; escalation ladders that terminate outside the venue's command structure.

6.3.2 Management $\times$ Circulation

What the cell measures. The human resources and procedures that manage in-venue crowd behaviour: circulation stewards, incident command structure, medical positioning, and response protocols for developing conditions.

Quantities used. Ratio of trained circulation stewards to occupancy; documented incident command hierarchy; medical positioning relative to sub-regions with elevated P_Ω ; and compatibility with the software monitoring layer.

Zone boundaries. Green corresponds to trained crowd management staff in all sub-regions of elevated P_Ω , with rehearsed incident command. Amber corresponds to general security coverage without dedicated crowd management specialisation. Red corresponds to absence of dedicated in-venue crowd management staff.

Typical failure modes. Reliance on general security without crowd-specific training; incident command that does not integrate with the software monitoring layer; medical positioning inconsistent with actual risk distribution.

6.3.3 Management $\times$ Egress

What the cell measures. The human resources and procedures that manage egress: egress stewards, dispersal coordination, transport liaison, and post-event sweep protocols.

Quantities used. Rehearsal history of egress procedures; documented dispersal coordination; transport-liaison protocol readiness; and compatibility with the egress hardware and software.

Zone boundaries. Green corresponds to rehearsed egress protocol with documented transport-liaison coordination. Amber corresponds to written but not recently rehearsed protocol. Red corresponds to absence of documented egress protocol beyond written signage.

Typical failure modes. Egress protocols that assume all exits will operate; dispersal coordination that does not account for transport-capacity constraints; sweep protocols that do not verify vacation of high-risk sub-regions.

6.4 Reading the Matrix

The nine cells of the venue P-report are populated during venue certification and updated periodically thereafter. Each cell carries a zone assignment (Green, Amber, or Red) and supporting documentation of the physical or organisational quantities that justify the assignment. A complete P-report includes not only the nine cell assignments but also the pairwise consistency verifications described in Section 5: for each of the three columns, the report documents that the Software cell is consistent with the Hardware cell, and that the Management cell is consistent with both.

The resulting document is auditable in the sense described in Section 3: a third party can verify the zone assignments against the supporting quantities, and can verify the consistency claims against direct observation of the venue. The report is not a narrative of what the venue does well; it is a structured representation of the venue’s static safety envelope, expressed in a form that supports insurance underwriting, regulatory certification, and—should the need arise—post-incident liability analysis.

7. Three-Zone Data Expression

Each cell of the venue P-report carries a zone assignment: Green, Amber, or Red. This section develops the physical basis for these zones, their boundaries, and the reasons for using a three-level ordinal expression rather than either a binary pass/fail or a continuous score.

7.1 Definition of the Three Zones

The three zones are defined by their relationship to the P-index framework (Tao, 2026a), which provides the physical basis for quantifying crowd compression risk. The P-index framework introduces two related quantities: P_{baseline} , the risk level under normal operating conditions, and P_{max} , the risk level under worst-case scenarios (panic triggered, directional convergence, extreme flow). The three zones correspond to well-defined regions in the joint space of these

two quantities.

Green zone. $P_{\text{baseline}} = 0$ under the operational scenarios relevant to the cell. This is a physical guarantee, not a statistical claim: under the modelled conditions, no individual accumulates above-threshold compression exposure, so no incident of the class that the P-index models can occur. The Green zone is the strong safety condition.

Amber zone. $P_{\text{baseline}} = 0$ but $P_{\text{max}} > 0$ under stress scenarios; or P_{baseline} takes values close to but not exceeding a specified operational limit. The Amber zone corresponds to conditions in which normal operation is physically safe but the margin against panic or extreme conditions is not itself guaranteed. Amber-zone cells require event-specific analysis to determine whether the event’s operational profile stays within the safe margin. This is where event-layer DIM ICE analysis provides the additional resolution.

Red zone. The reconstructed (F, τ) trajectory under P_{max} scenarios crosses the long-duration lethal anchor of the P-index framework (Tao, 2026a, Section 4.6), or P_{baseline} is non-zero under normal operational scenarios. Red-zone conditions indicate that the cell’s provisions are inadequate for events of any significant scale; incidents become physically inevitable given plausible triggering conditions. Historical compressive-asphyxia incidents (Hillsborough, Love Parade, Mecca, Astroworld, Itaewon) are consistent with this boundary as out-of-sample confirmation (Tao, 2026a, Section 6.3), but do not define it.

7.2 Physical Basis for Zone Boundaries

The zone boundaries are not arbitrary engineering thresholds. They correspond to physical transitions in the P-index framework, which in turn correspond to distinct regimes of the underlying contact mechanics.

The Green-Amber boundary corresponds to the transition of P_{baseline} from zero to non-zero. This is a physical phase transition in the sense described in Tao (2026a): below the transition, no individual accumulates above-threshold exposure regardless of the observation window; above the transition, at least some individuals do, and casualty risk becomes non-zero. The venue capacity limit N_{max} is defined precisely as the point of this transition.

The Amber-Red boundary corresponds to the point at which the reconstructed (F, τ) trajectory under P_{max} scenarios crosses the long-duration lethal anchor of the P-index framework. This anchor is calibrated independently of crowd incident data, from two evidence streams whose failure mechanism is purely respiratory restriction: sand-burial fatality series and forensic autopsy series of crush asphyxia (Tao, 2026a, Section 4.6). Applied to the five documented compressive-asphyxia incidents (Tao, 2026a, Section 6.3) as out-of-sample consistency checks, the framework yields P_{max} values in the range 2×10^{-4} to 5×10^{-3} . The Amber-Red boundary is not set from these values. It is defined by the anchor itself: a cell crosses into Red when its worst-case (F, τ) trajectory reaches the long-duration lethal anchor. The expression of this anchor as a venue-level P_{max} floor—of order 6×10^{-5} for a reference venue of 10^3 occupants observed over one hour, scaling with occupancy and observation window—is derived from the anchor definition and the P-index formula alone, independently of incident data (Tao, 2026a, Section 4.6). The five documented incidents all lie at or above this anchor-derived floor; this is an out-of-sample

consistency check—the incidents are found to have crossed the anchor at $P_{\max}$ values above the independently derived floor—not a calibration of the boundary to the incident set.

Neither boundary is fixed for all time. As the P-index framework’s calibration improves through accumulated commuter data (Tao, 2026a, Section 6.5), the numerical values of the boundaries will refine. The physical meaning of the boundaries—phase transition to non-zero P_{baseline} , and (F, τ) trajectory crossing the long-duration lethal anchor—remains fixed even as the numerical estimates evolve.

7.3 Why Three Zones Rather Than Alternatives

Several alternative representations were considered and rejected in the design of the framework.

Binary pass/fail. A binary representation would collapse the Amber zone into either Green or Red. Collapsing into Green would misrepresent the safety guarantee: Amber conditions are safe under normal operation but not under stress, and treating them as fully safe would mislead insurers and event organisers about the residual risk. Collapsing into Red would misrepresent the operational envelope: many venues that operate safely across many events sit in the Amber zone for at least some cells, and treating them as Red would foreclose legitimate operation. The Amber zone corresponds to a physically real intermediate condition that must be represented explicitly.

Continuous score. A continuous score would attempt to represent the full precision of the P-index values in each cell. This is technically feasible, but it introduces two problems. First, the P-index calibration is not currently precise enough to support meaningful discrimination at fine granularity; the joint uncertainty of the framework (Tao, 2026a, Section 6.4) is about half an order of magnitude. Continuous scoring would misrepresent this uncertainty by suggesting more precision than the underlying calibration supports. Second, continuous scores are harder to communicate to non-technical stakeholders and harder to defend in legal contexts, where discrete categories map more directly to standards of care.

Finer ordinal categories (five zones, seven zones, etc.). Finer categories multiply the boundary conditions that must be defended and increase the surface area for calibration disputes without adding decision-relevant resolution. Three categories align with the physical structure of the underlying framework (safe, marginal, unsafe), and this alignment is what gives the boundaries their physical rather than conventional character.

The three-zone expression is therefore chosen not as a compromise but as the representation that matches the physical granularity of the underlying framework and the decision granularity of the users.

7.4 Zone Assignment Procedure

For each cell of the venue P-report, zone assignment proceeds through the following steps:

1. Identify the physical or organisational quantities relevant to the cell, as specified in Section 6.

2. Determine the operational scenarios under which the cell will be evaluated: design load, stress load, and worst-case load.
3. Compute or estimate P_{baseline} and P_{max} under these scenarios, using the P-index framework methodology.
4. Compare the computed values against the zone boundaries specified for the cell.
5. Assign the zone (Green, Amber, or Red) that corresponds to the observed values.
6. Document the supporting quantities and the sensitivity of the assignment to reasonable variation in input assumptions.

The documentation supporting the zone assignment is as important as the assignment itself. A Green assignment without supporting quantities is not audit-defensible. The supporting documentation is what allows insurers, regulators, and—after an incident—forensic analysts to verify that the zone assignment was made rigorously and to identify what would change if the underlying conditions changed.

7.5 Zone Evolution Over Time

Zone assignments are not permanent. They reflect the state of the venue at the time of assessment, and they must be re-evaluated when material changes occur. Three types of change can shift a zone assignment.

Physical changes to the venue. Modifications to hardware—exit widening, corridor reconfiguration, permanent barrier placement—change the underlying physical quantities and therefore the zone assignment. Post-modification re-assessment is required.

Changes in supporting software or management. Even without hardware modification, changes to signage, monitoring, or staffing can shift a cell’s zone assignment. Software or management downgrades can move a cell from Green to Amber; upgrades can move it in the opposite direction.

Refinement of the underlying framework. As P-index calibration improves through accumulated data, the numerical boundaries of the zones may shift. A cell that was Green under early calibration may become Amber under refined calibration if the boundary tightens. This form of change is not a criticism of the earlier assessment; it reflects the monotonic improvement of the framework’s precision over time. Reports should be re-issued periodically to reflect current calibration.

The productive-capital framing developed in Section 9 treats these periodic re-assessments as part of the ongoing value of the P-report: each re-assessment maintains the report’s currency and, in the aggregate, contributes to the field’s calibration by adding another data point.

8. Complementary Usage of the Two Nine-Cell Reports

The venue P-report defined in Section 6 and the event DIM ICE analysis described in Section 2 share the horizontal axis (Ingress, Circulation, Egress) but differ on their vertical axis, their

scoring logic, and their object of analysis. This section develops how the two reports interface in practice, the scenarios in which each is used independently, and the scenarios in which they are used jointly.

8.1 The Baseline-and-Ceiling Interface

The venue P-report and event DIM ICE relate to each other through what we term the baseline-and-ceiling interface. Under this interface:

The venue P-report’s **Green zones** define the operational *floor* on which any event at the venue must operate. Green-zone cells represent physically-guaranteed safety margins that no event decision can improve upon in the short term (they are venue properties, not event properties) but that every event inherits by virtue of being held at that venue.

The venue P-report’s **Red zones** define the operational *ceiling* that no event may cross. Red-zone cells represent physical limits that no operational decision at the event layer can overcome. An event that requires operation in a Red-zone regime cannot safely be held at the venue, regardless of the sophistication of its DIM ICE analysis.

The venue P-report’s **Amber zones** define the *operating band* within which event-specific analysis is required. Amber cells describe conditions that are safe under normal operation but that carry residual risk under stress. Whether a specific event can safely operate within an Amber-zone cell depends on the event’s scale, audience, and management provisions—which is precisely what event DIM ICE analysis is designed to assess.

The interface between the two reports is therefore not a redundant overlap but a division of labour. The venue P-report answers the question: what safety envelope does this venue provide? The event DIM ICE answers the question: does this specific event fit safely within that envelope?

8.2 Independent Usage of the Venue P-Report

The venue P-report is used independently of any specific event in the following situations.

Venue licensing and certification. Regulatory bodies that certify venues for public occupancy require documentation of the venue’s static safety envelope. A completed venue P-report provides this documentation in a structured form that regulators can audit, compare across venues in their jurisdiction, and integrate with existing licensing frameworks.

Insurance underwriting. Insurers evaluating a venue for liability coverage require an assessment of the venue’s baseline risk independent of any specific event. The P-report’s zone assignments—particularly the identification of any Red-zone cells and the distribution of Amber-zone cells—directly informs the underwriting decision and the base premium calculation.

Venue rental marketing. A venue with a current, high-quality P-report can present its safety envelope to prospective event organisers as part of the rental proposition. Organisers selecting among candidate venues can compare P-reports and factor the venue’s baseline envelope into their own event planning.

Post-modification re-assessment. When a venue undergoes physical modification (structural

changes, exit reconfiguration, permanent barrier placement), the P-report is re-issued to reflect the modified envelope. The re-issued report becomes the reference document for subsequent events.

8.3 Independent Usage of the Event DIM ICE

The event DIM ICE is used independently of a formal venue P-report in the following situations.

Events at pre-certified venues. When an event is held at a venue whose P-report is current and whose zone assignments cover the operational regime of the planned event, the organiser's DIM ICE analysis can proceed without re-establishing the venue baseline. The DIM ICE focuses on the event-specific overlay: audience profile, programming, staffing plan, information systems, and contingency provisions.

Standard-scale events. For events whose scale and profile fall well within the venue's Green-zone envelope, the DIM ICE analysis is the primary safety documentation. The venue P-report exists as a background reference; the working document during event planning is the DIM ICE.

Legacy DIM ICE practice. The framework proposed here does not disturb existing DIM ICE practice. Organisers who have long produced DIM ICE analyses in the Still tradition continue to do so. The venue P-report is a complementary document that reduces the DIM ICE burden when a suitable venue P-report exists, but its absence does not prevent DIM ICE from being used in its established form.

8.4 Joint Usage for High-Attendance or High-Risk Events

The joint use of both reports is indicated in three classes of situations.

Events approaching venue capacity. When an event's expected attendance approaches the venue's $N_{\max}$ or when the event's density profile pushes any P-report cell from Green into the Amber band, joint documentation is required. The venue P-report establishes the baseline; the event DIM ICE demonstrates that the event's specific provisions maintain safe operation within the Amber band.

Events with elevated risk profiles. Events with heightened security profiles, politically sensitive events, or events with historically elevated incident rates (certain music genres, first-time festival formats, events with alcohol or substance risk factors) warrant joint documentation regardless of scale. The venue P-report demonstrates the physical baseline; the DIM ICE demonstrates the event-specific mitigation.

Events subject to enhanced regulatory scrutiny. Some jurisdictions apply enhanced review to events above defined thresholds. Joint documentation provides regulators with the complete safety analysis in a structured form suitable for expedited review.

8.5 Handoff Protocol

The practical handoff between venue P-report and event DIM ICE proceeds through the following steps.

1. The event organiser obtains the current venue P-report from the venue operator.
2. The organiser identifies which cells of the P-report are Green (inherited safety margins requiring no event-level analysis), Amber (requiring event-specific analysis), and Red (foreclosing certain event configurations).
3. The organiser’s DIM ICE analysis focuses on the Amber-zone cells. For each such cell, the DIM ICE documents the event’s specific provisions that maintain safe operation within the Amber envelope.
4. Cross-references between the two documents are maintained. Each Amber-zone cell in the P-report cites the corresponding DIM ICE section that addresses it; each DIM ICE section that addresses an inherited constraint cites the P-report cell that establishes it.
5. The two documents together constitute the joint safety analysis for the event.

The cross-referencing is not decorative. It creates a documentary trail that post-incident analysts can follow to reconstruct which layer was responsible for which aspect of the safety envelope. If an incident occurs at a Green-zone P-report cell, the incident implicates the venue baseline (or the P-report’s calibration). If it occurs at an Amber-zone P-report cell that the DIM ICE addressed, the incident implicates the event-specific provisions or their execution. If it occurs at an Amber-zone cell that the DIM ICE did not address, the incident implicates the organiser’s analytical completeness. This attribution structure supports the liability apportionment logic developed in Section 5 and applied in Section 10.

8.6 Neither Report Substitutes for the Other

A final observation on the relationship between the two reports: neither substitutes for the other. A rigorous venue P-report does not eliminate the need for event DIM ICE analysis, because the venue baseline does not capture event-specific dynamics. A rigorous event DIM ICE does not eliminate the need for a venue P-report, because event-layer analysis cannot verify the physical safety envelope of the venue it inherits.

This is analogous to the relationship between structural engineering certification of a building (a static property of the building) and occupancy-specific fire evacuation planning (a use-specific property overlaid on the building). The two forms of analysis address different questions and address them at different layers of the safety envelope. The proposal of this paper is to establish, for crowd-dense venues, the same clean separation between static baseline certification and dynamic event planning that has long been standard in structural and fire engineering.

9. Safety Reports as Productive Capital

9.1 The Cost-Center Framing and Its Limits

Safety reports in the crowd-dense venue sector are conventionally treated as compliance overhead. Venue operators produce them to satisfy insurers and regulators; event organisers produce

them to obtain permits. Under this framing, the incentive is to produce the minimum documentation that will pass review. Report quality is bounded from above by cost containment, and the effort invested in any single report has no economic afterlife once the specific compliance requirement is satisfied.

This framing is a legacy of the era in which safety documentation was fundamentally qualitative: a well-written report and a poorly-written report served the same compliance function, and there was no downstream mechanism through which additional investment in quality translated into additional value. Under the framework proposed in this paper, this is no longer the case. Quantified, physically-grounded, and cell-consistent reports generate downstream value that persists well beyond the specific event or certification they were produced for.

9.2 Venue Reports as an Asset

A complete venue nine-cell report—with each cell populated using the three-zone data expression and internal consistency verified through the Non-Contradiction Principle—becomes a persistent asset of the venue operator. Its value derives from several distinct channels.

Premium reduction. Insurers using the framework can price venue coverage based on the objective content of the report rather than on qualitative judgment. Venues with high-quality reports and favourable zone distributions attract lower base rates. This is the most direct and immediate value channel.

Rent uplift. A venue with a complete, current, high-quality report reduces the cost of every event that runs at it. Event organisers hosting events at such venues do not need to re-analyse the venue’s static safety envelope; they can inherit it and focus their DIM ICE effort on event-specific dynamics. This saved cost is transferable: the venue can capture a share of it through higher facility fees, and organisers still net out ahead because their total safety-analysis cost falls. High-quality venue documentation thereby becomes a competitive advantage in the venue rental market, not merely a compliance shield.

Event-mix expansion. Some categories of events—high-attendance concerts, international tournaments, events with heightened security profiles—are effectively closed to venues that cannot demonstrate a rigorous baseline safety envelope. A complete P-report opens the venue to event categories it would otherwise be excluded from, expanding its addressable market.

Litigation defence. In the event of an incident, a pre-existing complete report constitutes evidence that the venue operator met a documented standard of care at the time of the event. This does not immunise the operator from liability, but it materially shifts the burden and the likely apportionment.

Each of these channels produces recurring value across the useful life of the report. The report is not consumed by any single use.

9.3 Event Reports as a Reusable Template Library

A parallel logic applies to event organisers. A DIM ICE report produced for a specific event has traditionally been treated as bespoke work: produced under time pressure, filed after the

event, and rarely revisited except for post-incident review. Under the framework proposed in this paper, event DIM ICE reports become *templatable assets*.

An organiser who has produced a rigorous DIM ICE analysis for a mid-size outdoor music festival at a specific venue class can reuse the analysis structure for future events of similar type and scale, updating only the parameters that differ. This reduces the marginal cost of each subsequent event, and—more importantly—it allows the organiser to scale operations without a corresponding linear scale-up of safety-analysis costs.

For established multi-event operators (major concert promoters, festival networks, sporting-body event managers), a library of DIM ICE templates keyed to venue types, event scales, and audience profiles becomes a **capital asset that supports growth**. New events can be launched more rapidly and at lower risk because the safety analysis draws on validated prior work rather than being reconstructed each time.

This capability is particularly valuable for the emerging class of experience-based commercial activities—location-based entertainment, immersive events, seasonal festivals, brand activations—where speed to market is a competitive variable and safety analysis has historically been a bottleneck.

9.4 The Two-Sided Market Effect

The venue and event asset channels reinforce each other. A market in which a growing number of venues hold high-quality P-reports is a market in which event organisers can plan events more efficiently. A market in which a growing number of event organisers hold well-maintained DIM ICE template libraries is a market in which venues with high-quality baseline documentation are preferentially selected. Each side's investment in documentation quality raises the value of the other side's documentation.

This two-sided dynamic is the mechanism through which the framework, once adopted by a critical mass of participants, becomes self-reinforcing. Neither side needs to be persuaded that safety reports are worth investing in on abstract grounds. Each side observes that participants who invest in higher-quality documentation are systematically outperforming those who do not, on the measurable metrics that matter to them: insurance costs, rental income, event throughput, and post-incident exposure.

9.5 Implications for Framework Adoption

The productive-capital framing has a strategic consequence for how the framework enters the market. It does not need to be mandated by regulators before it becomes valuable. Its value to any individual venue or organiser is realised as soon as they adopt it, through the channels described above. Regulatory adoption, when it occurs, formalises and accelerates a dynamic that is already in motion in the commercial layer.

This ordering—commercial adoption first, regulatory formalisation second—is the same ordering that characterised the adoption of catastrophe modelling in the insurance sector during the 1990s. Firms that adopted physics-based catastrophe models before regulatory requirements

existed captured pricing advantages that persisted for a decade. The same dynamic is available in the crowd safety sector today.

10. Case Reconstruction: Hierarchy Violations in Historical Incidents

This section applies the framework developed in Sections 4 through 8 to four documented historical incidents. For each incident, we identify which cells of the venue P-report and event DIM ICE would have contained material provisions, which pairwise consistency checks (in the sense of the Non-Contradiction Principle) would have flagged the pre-incident risk, and which failure mode (Section 5) best characterises the incident.

We do not claim that a correctly executed nine-cell analysis would have prevented each incident with certainty. What we claim is that each incident, under the framework proposed here, would have been documented in a way that surfaced the specific pre-existing contradiction before the incident occurred. Whether that surfacing would have led to corrective action is a matter of institutional response beyond the framework's scope; but the framework's contribution is to make the contradiction visible in the first place.

The four cases are Hillsborough (1989), Love Parade (2010), Astroworld (2021), and Itaewon (2022). Detailed forensic reconstruction of these incidents from a physical standpoint is developed in the P-index framework paper (Tao, 2026a) and the anchor framework paper (Tao, 2026b). The present section focuses specifically on how the nine-cell framework would have represented each case.

10.1 Hillsborough, 1989: Management Contradicts Hardware

The Hillsborough disaster of 15 April 1989 occurred at the FA Cup semi-final at Sheffield Wednesday's Hillsborough Stadium and resulted in 97 fatalities. The direct physical mechanism was compressive asphyxia in the standing pens of the Leppings Lane terrace. The proximate operational cause, established through the Taylor Report and subsequent inquests, was the decision to open Gate C to relieve crowd pressure outside the stadium; this decision delivered a large additional volume into terrace pens whose capacity was already at or above safe limits.

Under the framework proposed here, the Hillsborough case exemplifies Failure Mode III: Management contradicts Hardware.

Hardware $\times$ Ingress and Circulation. The physical capacity of the Leppings Lane terrace pens was defined by their perimeter fencing and the physical space available. This capacity, in P-report terms, was fixed. The static P-baseline analysis of the terrace, executed rigorously, would have identified a physical capacity limit that any responsible operational decision was bound to respect.

Software $\times$ Ingress. The ingress signage and information systems directed supporters to the Leppings Lane end. Under the framework, the Software cell would document that this signage was designed for a specific ingress flow rate compatible with the hardware's physical capacity.

Management $\times$ Ingress. The critical decision was in this cell. The management response to ex-

ternal crowding—opening Gate C to admit a large additional volume rapidly—was inconsistent with the hardware capacity established in the Hardware cell of the same column.

Under the Non-Contradiction Principle, this contradiction would have been surfaced if the following pairwise check had been executed in advance: does the Management cell’s contingency protocol for external ingress pressure remain within the physical capacity documented in the Hardware cell? The answer, as post-incident analysis established, was no. The management protocol as it existed on the day contained no provision that limited large-batch admission by reference to internal capacity.

The framework does not claim that having executed this pairwise check would have altered the specific decisions on the day. What it claims is that the resulting documented contradiction would have been part of the pre-incident record, available to regulators, insurers, and—in the post-incident phase—to those apportioning responsibility. The Hillsborough Independent Panel’s findings, published in 2012, established that the same categories of failure were in fact identifiable in pre-existing documentation but were not surfaced by the analytic frameworks in use at the time.

10.2 Love Parade, 2010: Software Contradicts Hardware

The Love Parade disaster of 24 July 2010 occurred at the entry to the festival site in Duisburg, Germany, and resulted in 21 fatalities. The direct physical mechanism was compressive asphyxia in a tunnel-and-ramp bottleneck where bidirectional flow converged. The site had a single access tunnel serving both ingress and egress, with fixed signage directing convergent flows.

Under the framework proposed here, Love Parade exemplifies Failure Mode I: Software contradicts Hardware.

Hardware $\times$ Ingress and Egress. The physical geometry of the access tunnel and its downstream ramp was fixed and known. The tunnel’s throughput capacity was well below what would be required for the anticipated attendance under any bidirectional configuration.

Software $\times$ Ingress and Egress. The signage directed both ingress and egress through the same tunnel. The information layer, in other words, prescribed a use of the hardware that the hardware could not support without producing convergent flow.

Management $\times$ Ingress. Live management attempted to address the developing situation through crowd control at the tunnel mouth, but the fundamental incompatibility between the software prescription (bidirectional use of the tunnel) and the hardware capacity was structural.

The Non-Contradiction Principle, applied in advance, would have flagged the following pairwise check: is the Software cell’s flow prescription consistent with the Hardware cell’s capacity for the prescribed direction of use? The Love Parade site’s fixed signage prescribed a bidirectional flow through a tunnel whose hardware capacity could support only unidirectional flow at the anticipated volumes. This is exactly the class of contradiction that the framework is designed to surface.

Post-incident analysis of Love Parade, including the criminal proceedings that concluded without conviction in 2020, established that the incompatibility between the fixed infrastructure and

the flow plan was documented in engineering assessments produced before the event but not synthesised into a form that decision-makers could act upon. The framework proposed here would produce exactly such a synthesis, in the specific form of an Software-Hardware inconsistency flag on the venue P-report.

10.3 Astroworld, 2021: Management Contradicts Software

The Astroworld Festival crowd crush of 5 November 2021 occurred at NRG Park in Houston during a headline performance and resulted in 10 fatalities. The direct physical mechanism was compressive asphyxia in the crowd near the stage. Post-incident investigation identified that the event’s written safety protocol contained provisions for show suspension in response to crowd conditions, but the operational decision-making during the event did not execute these provisions in a timely manner.

Under the framework proposed here, Astroworld exemplifies Failure Mode II: Management contradicts Software.

Software × Circulation. The event’s written operational plan constituted the Software layer in the event-level DIM ICE sense. This plan included protocols for identifying crowd distress, communicating between security and production, and suspending the show if conditions warranted.

Management × Circulation. The live operational decisions during the event did not implement the written protocol at the point at which crowd conditions warranted show suspension. The management layer, in other words, did not execute what the software layer prescribed.

This is the paradigmatic case of Management contradicting Software: the written provisions were adequate in principle, but the live execution did not follow them. The Non-Contradiction Principle at the event layer would test not only whether the written protocol exists (the Software cell) but whether the operational structure (the Management cell) is configured to execute the protocol under the specific conditions the event will produce.

Astroworld also illustrates a distinct issue that the framework surfaces: the compatibility of the event’s Management cell with the venue’s Management cell. NRG Park’s baseline management provisions and Astroworld’s event-specific management provisions were, at the critical junctures, not integrated in a way that supported the required response. The handoff protocol described in Section 8—specifically the cross-referencing between venue and event Management cells—is designed to address exactly this class of failure.

10.4 Itaewon, 2022: Absence of Nine-Cell Analysis at Any Layer

The Itaewon crowd crush of 29 October 2022 occurred in a narrow sloped alley in the Itaewon district of Seoul during Halloween celebrations and resulted in 159 fatalities. The direct physical mechanism was compressive asphyxia under bidirectional flow with deep force chains. No formal organiser existed for the gathering: it was an emergent street-scale event drawing on the district’s Halloween tradition rather than a licensed event with defined attendance.

Under the framework proposed here, Itaewon exemplifies a distinct failure category: the absence of any nine-cell analysis at either the venue or event layer.

No venue P-report. The alley was not a licensed event venue and had no P-report. Its static safety envelope—critically, the physical capacity of the sloped alley under bidirectional flow—was not documented anywhere in a form that would have made its Red-zone character visible before the event.

No event DIM ICE. No organiser existed to produce a DIM ICE. The gathering emerged from social coordination without an entity assuming safety analysis responsibility. Municipal authorities did not classify the anticipated gathering as an event requiring safety documentation.

The Itaewon case highlights a limitation of the framework proposed here: it applies to venues and events with identifiable operators and organisers. Where no such entities exist, the framework provides no direct analytic tool. But it also identifies a distinct policy implication: the absence of a designated organiser for a foreseeable gathering is itself a condition that regulators can identify in advance, and the framework provides a structured form (the venue P-report for street-scale sites with recurring gatherings) that municipalities can produce for their own gathering-prone spaces.

The four cases together illustrate the three canonical failure modes formalised in Section 5, plus the null case of no analysis at any layer. Each failure mode maps to a specific class of pre-existing contradiction that the framework’s pairwise consistency check would have surfaced. What varies across the cases is the layer at which the contradiction originated and the party whose action or omission produced it—which is precisely the information required for post-incident liability apportionment.

11. Limitations and Discussion

This section addresses the scope limitations of the framework, the conditions under which the framework’s outputs are and are not reliable, and the relationships between the framework’s claims and broader considerations in professional practice, regulation, and law.

11.1 The Framework Structures Judgment; It Does Not Replace It

The nine-cell framework, and the Non-Contradiction Principle within it, provides a structure within which safety analysis proceeds. It does not eliminate the need for expert judgment at any stage. Populating a cell of the venue P-report requires expertise in the physical, informational, or operational domain of that cell. Determining whether the content of two cells is consistent requires understanding of how the described provisions would interact in practice. Assigning a cell to a zone requires calibrated judgment about the P-index values implied by the cell’s content and the operational scenarios relevant to it.

What the framework provides is a structure that directs expert judgment to the points at which it is most consequential: the boundaries between zones, the interactions between cells, and the joint safety envelope that the report as a whole describes. Judgment applied to these points has more leverage than judgment applied to filling in checklist items, which is the mode that conventional nine-cell reports encourage.

A related limitation is that the framework does not automate compliance. Producing a rigorous

venue P-report or event DIM ICE remains a professional exercise requiring trained personnel. The framework improves the ratio of consequential decision points to routine documentation, but it does not reduce the total professional labour required for high-quality safety analysis.

11.2 Zone Boundary Precision Depends on Underlying P-Index Calibration

The three-zone data expression developed in Section 7 inherits its numerical precision from the P-index framework (Tao, 2026a). The P-index framework is currently calibrated against a limited set of public crowd-dynamics datasets, with historical incidents used as out-of-sample consistency checks rather than calibration targets (Tao, 2026a, Sections 6.3, 6.5); its joint uncertainty is approximately half an order of magnitude in the $\log P_{\max}$ range.

This uncertainty has direct implications for the zone assignments. Cells whose underlying P-index values are far from any zone boundary are unambiguously classified. Cells whose values are near a boundary may be ambiguous under current calibration, and the classification in such cases carries an explicit uncertainty that the report should document. As the P-index framework's calibration improves through the commuter-data reverse calibration route described in Tao (2026a, Section 6.5), the ambiguous band around each boundary narrows. The framework is therefore intrinsically improvable: its precision is not fixed at its current level.

An implication is that P-reports carry a calibration date. A P-report issued under early calibration and re-issued under refined calibration may show different zone assignments for cells near boundaries, even if the underlying physical venue has not changed. This is not a defect of the framework but a normal consequence of calibrated analysis. Reports should document the calibration version under which they were produced.

11.3 Legal Admissibility Varies by Jurisdiction

The framework provides physical and organisational analysis of a form usable in insurance underwriting, regulatory certification, and civil litigation. Whether the outputs of such analysis are legally admissible in any specific proceeding is a question of jurisdictional law, not of the framework itself. Common-law jurisdictions with established expert-witness traditions and case law on tort liability tend to accept quantitative safety analysis of this form more readily than jurisdictions with more restrictive evidentiary rules.

The framework is designed to produce outputs that meet standards of physical rigor and methodological transparency that any responsible admissibility review would accept. Whether any specific court in any specific jurisdiction actually accepts them is beyond the scope of what the framework can guarantee.

An observation that follows from this: the framework's admissibility improves through case-law accumulation. Each successful use of the framework in a legal proceeding establishes precedent for subsequent uses. The framework's legal utility is therefore expected to grow non-linearly with adoption, particularly in jurisdictions where the first several successful uses establish the framework as recognised expert methodology.

11.4 Internal Consistency Does Not Guarantee External Adequacy

The Non-Contradiction Principle establishes a necessary condition for safety: the three cells in any column must be mutually consistent. It does not establish a sufficient condition. A venue whose nine cells are all mutually consistent may nonetheless describe an inadequate safety strategy if all cells are jointly under-specified.

For example: a venue whose Hardware cell describes minimal egress capacity, whose Software cell describes signage consistent with that minimal capacity, and whose Management cell describes staffing consistent with the software, may pass the internal-consistency check while nonetheless being inadequate for the events actually held there. The consistency check verifies that the three cells describe a coherent strategy; it does not verify that the strategy is adequate to the venue's actual use profile.

Adequacy is addressed by the zone assignments. A cell in the Red zone is inadequate regardless of whether it is internally consistent with the other cells in its column. A venue with even one Red-zone cell is not certified for events that would engage that cell, regardless of the consistency of the overall report. The framework's two analytic tools—the three-zone assignments (within-cell adequacy) and the Non-Contradiction Principle (between-cell consistency)—are jointly necessary. Neither alone is sufficient.

11.5 Interaction with Existing Regulatory Frameworks

The framework proposed here is designed to interact additively with existing regulatory frameworks, not to replace them. Fire codes, per-person area regulations, structural engineering certifications, and evacuation-time analyses continue to apply. The venue P-report supplements these with a compressive-asphyxia-focused safety envelope that existing frameworks do not directly address.

In jurisdictions where regulatory bodies adopt the framework formally, the venue P-report becomes part of the required documentation for venue certification. In jurisdictions where adoption is initially commercial (insurance-driven, market-driven), the P-report exists alongside required regulatory documentation without replacing it. The productive-capital framing developed in Section 9 does not require regulatory adoption to generate value.

An implication for regulators considering adoption: the framework is compatible with existing regulatory documentation because it addresses a domain (dynamic compressive-asphyxia risk) that existing frameworks do not cover. Adoption does not require dismantling existing requirements. This compatibility is a design choice, not an accident.

11.6 Scope Boundaries of the Framework

Finally, several classes of situation lie outside the framework's scope and are not addressed by the analysis developed here.

Non-compressive crowd incidents. The P-index framework addresses compressive-asphyxia risk specifically. Crowd incidents driven by other mechanisms (fire, structural collapse, active-threat

scenarios, mass poisoning) require separate analytic frameworks. The venue P-report does not purport to cover these mechanisms.

Emergent gatherings without designated organisers. As discussed in the Itaewon case (Section 10), the framework applies to venues and events with identifiable operators. Where no such entity exists, the framework provides no direct tool. The Itaewon-type policy implication—municipalities producing P-reports for their own gathering-prone spaces—is a partial response to this limitation but does not fully close it.

Individual medical incidents. Crowd density can contribute to individual medical emergencies (cardiac events, heat illness, drug reactions) that are not primarily driven by compressive mechanics. The framework addresses the compressive-asphyxia baseline; the broader class of density-associated medical risk requires separate consideration.

These scope boundaries do not diminish the framework’s utility within its designated domain. They clarify what the framework claims to provide and what it does not.

12. Conclusion

This paper has proposed a bipartite nine-cell framework for crowd safety analysis. The framework consists of two structurally analogous 3×3 matrices: a venue P-report and an event DIM ICE analysis. The two matrices share the horizontal axis (Ingress, Circulation, Egress) but differ on their vertical axis, their scoring logic, and their object of analysis.

12.1 Recapitulation of Contributions

Five principal contributions have been developed in the paper.

The bipartite structure. The framework separates venue-layer analysis (static, physical, long-lived) from event-layer analysis (dynamic, operational, event-specific). This separation clarifies the division of analytic labour between venue operators and event organisers, and it clarifies the domains within which each party’s responsibility applies.

The Hardware/Software/Management vertical axis. The venue P-report’s vertical axis is ordered by cost of change: Hardware is the layer with the highest cost to modify, Software the middle layer, Management the layer at which real-time adaptation is possible. This ordering, made explicit, transforms the vertical axis from a set of parallel domains into a dependency stack in which higher layers must respect the constraints of the layers below.

The Non-Contradiction Principle. Within each column of the venue P-report, the three cells must be mutually consistent: Software must not contradict Hardware, and Management must not contradict either. The three canonical failure modes—Software contradicting Hardware, Management contradicting Software, and Management contradicting Hardware—correspond to identifiable classes of historical incident and to identifiable classes of pre-existing documentary contradiction that the framework’s pairwise consistency check surfaces.

The three-zone data expression. Each cell of the venue P-report carries a zone assignment (Green, Amber, or Red) whose boundaries correspond to well-defined physical phase transitions

in the underlying P-index framework. The three-zone expression is chosen to match the physical granularity of the underlying analysis and the decision granularity of the users.

The productive-capital framing. Safety reports produced under this framework are treated not as compliance overhead but as commercial assets that generate downstream value through premium reduction, rent uplift, event-mix expansion, and litigation defence. This framing changes the economic logic of safety documentation and provides a commercial pathway for framework adoption independent of regulatory mandate.

12.2 Positioning Relative to Existing Practice

The framework is positioned to complement, not to replace, existing crowd safety tools. DIM ICE remains the appropriate framework for event-layer analysis, and the paper preserves Still’s DIM ICE structure unchanged in this domain. Fire codes and per-person area regulations remain applicable to their intended domains. The physical basis of the framework’s quantitative content is provided by the P-index framework (Tao, 2026a), and its forensic application draws on the anchor framework (Tao, 2026b).

The framework’s contribution is to fill a specific gap: venue-layer static safety certification with a compressive-asphyxia focus, integrated into a structure that interfaces cleanly with existing event-layer practice. This gap has been identifiable in the professional practice literature for at least a decade, and the framework proposed here represents one attempt to close it.

12.3 Path to Adoption

The productive-capital framing developed in Section 9 implies a specific adoption sequence: commercial value channels first, regulatory formalisation second. Venue operators and event organisers who adopt the framework realise value through insurance, rental, and litigation-defence channels before any regulatory mandate exists. Insurers who adopt the framework realise value through improved underwriting precision. Legal practitioners who adopt the framework realise value through improved evidentiary chains in liability proceedings.

Regulatory adoption, when it occurs, formalises and accelerates a dynamic that is already in motion in the commercial layer. This ordering—commercial adoption preceding regulatory formalisation—is the same ordering that characterised the adoption of physics-based catastrophe modelling in the insurance sector during the 1990s, and there is no structural reason it should not characterise the adoption of physics-based crowd safety analysis in the decade ahead.

12.4 Companion Papers and Forward Direction

The framework proposed in this paper is one of a series of related contributions developed under the PneumaTheorem programme. The P-index framework (Tao, 2026a) provides the physical basis for the venue P-report’s quantitative content. The anchor framework (Tao, 2026b) provides the forensic methodology for reconstructing historical incidents in a form usable in legal and insurance contexts. Additional forthcoming papers will address specific case studies in depth and will develop the technical protocols for report production.

The framework, taken as a whole, offers a physically grounded, structurally coherent, and commercially adoptable path forward for crowd safety analysis. Its adoption is not guaranteed; the value channels it activates depend on individual actors in venues, events, insurance, and regulation making adoption decisions in the years ahead. What the framework provides is the analytic substrate on which those decisions can be made. The rest is a matter of practice.

Data and Code Availability

This paper is a methodological specification and does not analyse primary experimental data. The historical incident cases discussed in Section 10 (Hillsborough 1989, Love Parade 2010, Astroworld 2021, Itaewon 2022) are drawn from publicly available investigation reports, forensic literature, and official inquiries cited in the References and in the companion physical framework paper (Tao, 2026a). The framework’s numerical basis derives from the P-index framework (Tao, 2026a) and the anchor framework (Tao, 2026b); readers seeking quantitative reproducibility should consult those companion papers. Reference cell templates, worked P-report examples, and consistency-check protocols will be published separately under a persistent Zenodo DOI. No proprietary software accompanies this specification-based framework; any implementation satisfying the specifications of Sections 4 through 7 constitutes a valid backend.

Conflict of Interest

The author is the founder of PneumaTheorem, a research entity engaged in crowd-risk analysis. The framework is open-access under CC BY 4.0; commercial services are offered separately through PneumaTheorem. This paper is not a commercial engagement, expert-witness statement, or regulatory certification.

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